

Minimizing irreversible losses in quantum systems by local counterdiabatic driving

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Counterdiabatic driving protocols have been proposed [Demirplak M, Rice SA (2003) *J Chem Phys A* 107:9937–9945; Berry M (2009) *J Phys A Math Theor* 42:365303] as a means to make fast changes in the Hamiltonian without exciting transitions. Such driving in principle allows one to realize arbitrarily fast annealing protocols or implement fast dissipationless driving, circumventing standard adiabatic limitations requiring infinitesimally slow rates. These ideas were tested and used both experimentally and theoretically in small systems, but in larger chaotic systems, it is known that exact counterdiabatic protocols do not exist. In this work, we develop a simple variational approach allowing one to find the best possible counterdiabatic protocols given physical constraints, like locality. These protocols are easy to derive and implement both experimentally and numerically. We show that, using these approximate protocols, one can drastically suppress heating and increase fidelity of quantum annealing protocols in complex many-particle systems. In the fast limit, these protocols provide an effective dual description of adiabatic dynamics, where the coupling constant plays the role of time and the counterdiabatic term plays the role of the Hamiltonian.

counterdiabatic driving | adiabatic gauge | transitionless driving | variational principle | complex systems

Despite the time-reversal symmetry of the microscopic dynamics of isolated systems, losses are ubiquitous in any process that tries to manipulate them. Whether it is the heat produced in a car engine or the decoherence of a qubit, all losses arise from our lack of control on the microscopic degrees of freedom of the system. Since the early days of thermodynamics and actually, even before, the adiabatic process has emerged as a universal way to minimize losses, leading to the concept of Carnot efficiency—the cornerstone of modern thermodynamics. Despite its conceptual importance, practical implications of the Carnot efficiency are limited, because the maximal efficiency goes hand in hand with zero power. Nonetheless, by sacrificing some of the efficiency, one can run the same Carnot cycle at finite power (1). Heat engines might seem to be a problem of the past, but the understanding of finite-time thermodynamics in small (quantum) systems has become increasingly important because of developments in quantum information and nanoengineering.

Developing and understanding methods to induce quasiadiabatic dynamics at finite times are paramount to the advancement of quantum information technologies. In general, one could distinguish between two (complementary) approaches. On one hand, one can, for a fixed setup, try to develop optimal driving protocols that result in minimal loss under certain constraints. Such protocols were recently suggested as appropriate geodesic paths in the parameter space in both the context of thermodynamics (2) and the context of adiabatic-state preparation (3, 4). The optimal protocols were also analyzed numerically using various optimum control ideas (5–9). On the other hand, one can try to engineer fast nonadiabatic protocols that lead to the same result as the fully adiabatic protocol. In particular, transitionless driving protocols were recently proposed and explored in small single-particle systems (10–15) with numerical extensions

to larger interacting systems (16). In this approach, one introduces an auxiliary counterdiabatic (CD) Hamiltonian drive on top of a target Hamiltonian to suppress all transitions between eigenstates. A general problem with this approach is that, in complex chaotic systems, the exact CD Hamiltonian is nonlocal and exponentially sensitive to any tiny perturbations. The goal of this work is to overcome these difficulties by providing a route to finding approximate optimal CD driving protocols by restricting to a class of physical operators (e.g., those accessible in experiments). In this work, we focus specifically on local CD protocols.

The ideas of CD driving are certainly not new and are used on an everyday basis in nature. Let us illustrate these ideas using an example of a waiter bringing a tray with a glass of water from the bar to a table (Fig. 1). As we will show in this work, this simple example contains very important insights, which will become relevant to the core of this paper. The goal of the waiter is to deliver the water to the table with a high fidelity (i.e., without spilling or splashing it). Of course, the glass should be vertical in the beginning of the process (i.e., when the waiter is leaving the bar) and at the end of the process (i.e., when the waiter reaches the table). The simplest protocol that can be adopted by the waiter is adiabatic, where he slowly moves along the shortest path (geodesic) connecting the bar and the table, keeping the tray vertically at all times. This protocol will work but will require a lot of time, and thus, the efficiency of such an “adiabatic waiter” will be very low. An efficient waiter has to serve more customers by going faster, and doing so requires a different tactic. When accelerating to reach a finite speed, a pseudoforce will act on the drinks, which will cause the drinks to spill or even tip over. This effect can be avoided by acting on the drinks with an equal and opposite force, which is exactly what waiters do. Moreover, the same tilt can counter a drag force caused by the air if the waiter runs very

Significance

Losses are ubiquitous in manipulating complex systems. They arise from our lack of control on the microscopic degrees of freedom of the system. A universal way to minimize losses is to consider adiabatic processes. These processes are, however, very slow, which significantly limits their power. In this work, we show how to speed up these protocols for general complex (quantum) systems. Although dissipation cannot be avoided, we show how it can be reduced significantly with only local access to the system. Applications range from quantum information technologies to preparing experiments and even controlling complicated classical systems, such as those found in nature.

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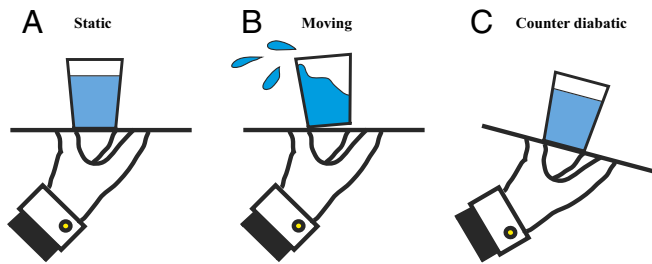


Fig. 1. The CD waiter. (A) A waiter's goal is to deliver a tray with a glass of water from the bar to a customer without spilling. In the beginning and the end of the task, the system should look like the situation in A. An adiabatic waiter can always be in the situation in A, but with a desire to be more efficient and speed up the protocol, a naive waiter will find himself in the undesirable situation in B somewhere during the task. (C) By tilting the tray, an example of CD driving, the situation in B can be avoided, and the desired tasks can be achieved much faster.

fast. In fact, by tilting the tray while moving, the waiter induces a CD force. Despite the fact that the system of the tray, the glass, and water is complex and chaotic, it is clear from our everyday experience that this CD protocol can be extremely efficient. Let us highlight several important points, which we can learn from this intuitive example. We will come back to these points later when we discuss various physical examples.

To implement an efficient CD protocol, one has to introduce new degrees of freedom (like a tilt), which do not show up in the initial and final states as well as the adiabatic path.

The system does not generally follow an instantaneous ground state: at intermediate times, the waiter tilts the tray and moves it fast, which correspond to a highly excited state of the system in the laboratory frame.

The CD protocol corresponds to adding local terms to the Hamiltonian of the system, like the gravitational field. This protocol is only sensitive to the velocity and acceleration of the waiter.

As we will show, these observations underlie crucial ideas behind engineering CD protocols in complex systems, such as locality and gauge equivalence. Using these ideas as a guiding principle, we develop a simple variational approach allowing one to find local and robust approximate counteradiabatic Hamiltonians. These counterterms allow one to achieve truly spectacular results in suppressing heating or targeting ground states of gapped or gapless many-particle systems with a very high fidelity at very fast speeds. An important advantage of the variational method is that it allows one to find efficient CD protocols without the need of diagonalizing the Hamiltonian, in particular, in the thermodynamic limit. Moreover, one can check the accuracy of the variational ansatz by analyzing the stability of the protocol with respect to adding additional terms.

Local CD Driving

CD Driving in Quantum and Classical Systems. Let us have a closer look at how transitions between eigenstates actually arise and how one can suppress them. Consider a state $|\psi\rangle$ evolving under the Hamiltonian $H_0(\lambda(t))$, which is time-dependent through the parameter $\lambda(t)$. In general, λ can be a multicomponent vector parameter (for example, in the case of a waiter, λ can stand for his x and y coordinates), but in this work, we will focus on the single-component case to avoid extra complications. If the parameter changes in time, then for a moving observer in the instantaneous eigenbasis of H_0 , the laws of physics are modified. This phenomenon is, of course, very well known for the case of an accelerated or rotating frame, but in fact, it applies to all types

of motion. Specifically, the Hamiltonian picks up an extra contribution and becomes

$$H_0^{\text{eff}} = \tilde{H}_0 - \dot{\lambda} \tilde{\mathcal{A}}_\lambda; \quad [1]$$

Here, $\tilde{\mathcal{A}}_\lambda$ is the adiabatic gauge potential in the moving frame. It is geometric in origin and related to the infinitesimal transformations of the instantaneous basis states in the quantum case and the infinitesimal canonical transformations of conjugate variables (like coordinates and momenta) in the classical case (details are in *Methods* and ref. 17).

In the moving frame, the Hamiltonian $\tilde{H}_0$ is diagonal (stationary), and therefore, all nonadiabatic effects must be caused by the second term. The idea of the CD driving is to evolve the system with the Hamiltonian

$$H_{\text{CD}}(t) = H_0 + \dot{\lambda} \mathcal{A}_\lambda,$$

such that, in the moving frame, $H_{\text{CD}}^{\text{eff}}(t) = \tilde{H}_0$ is stationary and no transitions occur. Note that, by construction in the zero velocity limit $|\dot{\lambda}| \rightarrow 0$, the CD Hamiltonian $H_{\text{CD}}(t)$ reduces to the original Hamiltonian $H_0(t)$ as expected. It is easy to show that the gauge potential $\mathcal{A}_\lambda$ satisfies the following equation (details are in *Methods* and ref. 17):

$$[i\hbar\partial_\lambda H_0 - [\mathcal{A}_\lambda, H_0], H_0] = 0. \quad [2]$$

This equation immediately extends to classical systems by replacing the commutator with the Poisson brackets: $[\dots] \rightarrow i\hbar\{\dots\}$. Although in this work, we focus on quantum systems, all general results and methodology equally apply to classical systems.

One can immediately see that Eq. 2 reduces to familiar Galilean transformations in the case of translations. Let us assume that

$$H_0 = \frac{p^2}{2m} + V(q - \lambda(t)), \quad [3]$$

that is, $\lambda(t)$ is the position of the center of the potential. In this case, the gauge potential is just the momentum operator $\mathcal{A}_\lambda = p$. Indeed, we have $i\hbar\partial_\lambda H_0 \equiv -i\hbar\partial_q V = -i\hbar[\partial_q, H_0] = [p, H_0]$, such that Eq. 2 is automatically satisfied. Thus, the exact CD Hamiltonian is

$$H_{\text{CD}} = H_0 + \dot{\lambda} p. \quad [4]$$

If the waiter implements the protocol (Eq. 4), then he would be able to move a tray with glasses fast without exciting it. Notice that this is not what the waiter actually does. The reason is that it is very hard to realize the term linear in momentum. Moreover, from Eq. 2, we see that, in systems satisfying time-reversal symmetry, where the Hamiltonian H_0 is real, the gauge potential and hence, the corresponding CD terms are always strictly imaginary. Therefore, the CD driving always breaks the time-reversal symmetry. The situation is actually much better than it seems. The CD term in this setup plays the role of the vector potential in electromagnetism:

$$\begin{aligned} H_{\text{CD}} &= \frac{p^2}{2m} + V(q - \lambda) + \dot{\lambda} p \\ &= \frac{(p + m\dot{\lambda})^2}{2m} + V(q - \lambda) - \frac{m\dot{\lambda}^2}{2}. \end{aligned} \quad [5]$$

It is very well-known that, if the vector potential is curl-free, which is always the case in one dimension, it can be removed (gauged away) at the expense of introducing a scalar potential via $p \rightarrow p + \partial_q f$ and $H \rightarrow H + \partial_t f$ for an arbitrary function $f(q, t)$. Choosing $f(q, t) = -m\dot{\lambda}q$, we see that

$$H_{\text{CD}} \sim \frac{p^2}{2m} + V(q - \lambda) - m\ddot{\lambda}q. \quad [6]$$

Thus, as expected from the Galilean invariance, the CD driving amounts to adding an extra gravitational field proportional to the acceleration. Here, we used the tilde sign instead of an equal sign

to highlight that the right-hand side is gauge-equivalent rather than equal to the Hamiltonian (Eq. 4). There is an important physical difference between the two CD protocols. Although following the imaginary CD protocol (Eq. 4) amounts to instantaneously following eigenstates of H_0 , following the real CD protocol (6) amounts to instantaneously following eigenstates of a gauge-equivalent Hamiltonian. Only when the velocity $\dot{\lambda}$ is zero do the two Hamiltonians coincide, such that CD driving leads to identical results. This subtlety is precisely the reason why the waiter, following the real CD protocol, does not keep the glass in the instantaneous ground state, except when the velocity and the acceleration are zero.

Although Eq. 2 is linear and looks very simple, this simplicity is actually deceptive. In fact, one can show that, in generic chaotic systems, it has no solution. In quantum chaotic systems, the exact analytic expression for $\mathcal{A}_\lambda$ suffers from the problem of small denominators (17), and in classical chaotic systems, it can be expressed through a formally divergent integral (18). The physical reason behind is very intuitive. By trying to find the exact gauge potential, we are requiring too much. We essentially want to find a transformation that keeps the system in exact many-body eigenstates without any excitations. However, achieving this is clearly impossible or at least exponentially hard [e.g., in chaotic many-particle systems, eigenstates are essentially random vectors sensitive to exponentially small perturbations of the Hamiltonian (19), and therefore, the exact gauge potential should have the same exponential sensitivity to the details of the many-body spectra and access to all microscopic degrees of freedom. However, finding such gauge potential is hardly our goal either. We are generally interested in either suppressing dissipation (i.e., suppressing transitions between levels resulting in substantial energy changes) or following very special states like the ground states, which are also robust to small perturbations. Thus, our goal should be finding approximate gauge potentials, which satisfy requirements of robustness and locality and strongly suppress physical diabatic effects rather than completely eliminate them, which is precisely what we are going to discuss next.

Variational Principle and Local CD Protocols. Our goal is to set up a variational procedure allowing one to determine the best possible $\mathcal{A}_\lambda$ under some constraints, like locality, robustness, or just experimental accessibility. Unconstraint minimization should, of course, result in the exact gauge potential (Eq. 2). It is easy to show that solving Eq. 2 is equivalent to minimizing the Hilbert-Schmidt norm of the operator

$$G_\lambda(\mathcal{A}_\lambda) \equiv \partial_\lambda H_0 + \frac{i}{\hbar} [\mathcal{A}_\lambda, H_0]$$

with respect to $\mathcal{A}_\lambda$. Indeed, finding the minimum of this norm is equivalent to the Euler-Lagrange equation,

$$\frac{\delta S(\mathcal{A}_\lambda)}{\delta \mathcal{A}_\lambda} = 0, \quad [7]$$

of the action (*Methods*)

$$S(\mathcal{A}_\lambda) = \text{Tr} [G_\lambda^2(\mathcal{A}_\lambda)]. \quad [8]$$

For classical systems, trace should be replaced by an integral over the phase space. Instead of minimizing the action over the whole Hilbert space of operators, one can now restrict to a subspace of physical operators. Let us introduce $\mathcal{A}_\lambda^*$ as an approximate adiabatic gauge potential; then, one can simply calculate the best constrained approximation by minimizing over all allowed $\mathcal{A}_\lambda^*$. To do so, one simply has to be able to evaluate the action (Eq. 8) for the allowed operators $\mathcal{A}_\lambda^*$. Evaluating the trace of local operators or their products is very straightforward and can be usually done with minimal efforts. Physically, the action (Eq. 8) defines the average transition rate (over all possible states) when the classical parameter $\lambda(t)$ is a weak random white noise pro-

cess. Therefore, minimizing the action is equivalent to suppressing processes, such as heating and energy diffusion, under a white noise drive.

Quite often, one is interested in suppressing transitions from a low-temperature manifold of states, in particular, the ground state. Then, targeting the gauge potential, which suppresses transitions everywhere in the spectrum, is overdemanding. Instead, one can define the action through a finite temperature norm:

$$S(\mathcal{A}_\lambda^*, \beta) = \langle G_\lambda^2(\mathcal{A}_\lambda^*) \rangle - \langle G_\lambda(\mathcal{A}_\lambda^*) \rangle^2, \quad [9]$$

where the angular brackets denote usual average with respect to the thermal density matrix $\rho(\beta) = 1/Z \exp[-\beta H_0]$. The Hilbert-Schmidt norm (Eq. 8) is recovered as the infinite temperature limit ($\beta \rightarrow 0$) of this norm [note that subtracting the second term in Eq. 9 is not affecting the result, because $\langle G_\lambda(\mathcal{A}_\lambda^*) \rangle$ is independent of $\mathcal{A}_\lambda^*$]. In the zero temperature limit $\beta \rightarrow \infty$, the action (Eq. 9) reduces to the variance of G_λ in the ground state. The exact gauge potential minimizes the action (Eq. 8) for any temperature. However, details of the variational solution can depend on β . In this work, we are focusing on finding CD protocols minimizing the infinite temperature action (Eq. 8), leaving the analysis of the finite/zero temperature action for future work.

Applications

This section is primarily concerned with applications in quantum mechanics; for convenience, we set $\hbar = 1$. At the end of the section, we show how to apply the current variational method to classical systems.

1D Lattice Fermions. Let us investigate the performance of variational CD protocols using an example of noninteracting lattice spinless fermions in a time-dependent potential. Despite the simplicity of the setup, the problems that we will be addressing are highly nontrivial, because we will consider inserting and moving obstacles, breaking translational symmetry. In turn, this perturbation generally leads to strong mixing of single-particle orbitals and strong nonadiabatic effects originating from scattering of fermions from the obstacle. Consider a single-band tight binding model in an external potential

$$H_0 = -J \sum_{j=1}^{L-1} (c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j) + \sum_{j=1}^L V_j(\lambda) c_j^\dagger c_j, \quad [10]$$

where $c_j^\dagger$ and c_j are fermionic creation and annihilation operators, respectively, and $V_j(\lambda)$ is the external potential. In principle, the dependence of V on λ can be arbitrary. Here, we will focus on a particular example of inserting the potential: $V_j(\lambda) = \lambda v_j$. In *SI Text*, we additionally analyze a moving potential: $V_j(\lambda) = V(j - \lambda)$. Because the Hamiltonian is real, as we discussed earlier, the adiabatic gauge potential should be purely imaginary [which is also clear from the form of the action (Eq. 8)]. The most local Hermitian imaginary operator that one can find for a free system is the current; therefore, we will look for solutions of the form

$$\mathcal{A}_\lambda^* = i \sum_{j=1}^{L-1} \alpha_j (c_{j+1}^\dagger c_j - c_j^\dagger c_{j+1}).$$

Substituting this potential into the action (Eq. 8) and extremizing with respect to the coefficients α_j , we find the following equation (*SI Text*):

$$-3J^2 \partial_j^2 \alpha_j + (\partial_j V_j(\lambda))^2 \alpha_j = -J \partial_j \partial_\lambda V_j(\lambda), \quad [11]$$

where $\partial_j \alpha_j \equiv \alpha_{j+1} - \alpha_j$, $\partial_j V_j \equiv V_{j+1} - V_j$, and $\partial_j^2 \alpha_j \equiv \alpha_{j+1} - 2\alpha_j + \alpha_{j-1}$ are discrete lattice derivatives. If the potential is smooth on the scale of the lattice spacing, then the discrete derivatives can be substituted by continuous derivatives. In general expression Eq. 11 is a set of linear equations, which can

always be solved numerically. However, there are several cases where one can find approximate or exact analytic solutions.

The imaginary hopping terms in $\mathcal{A}_\lambda^*$ explicitly break time-reversal symmetry, which might create some difficulties in implementing them in practice. However, restricting the variational ansatz only to the nearest neighbors allows one to once again perform a simple gauge transformation similar to the vector potential shift, also known as the Peierls substitution. Namely, $c_j \rightarrow c_j e^{-if_j}$, such that the new Hamiltonian becomes real (more details are in *SI Text*):

$$H_{CD} \sim - \sum_j J_j \left(c_{j+1}^\dagger c_j + h.c. \right) + \sum_j U_j c_j^\dagger c_j \quad [12]$$

with

$$J_j = J \sqrt{1 + (\dot{\lambda} \alpha_j / J)^2}, \quad [13]$$

$$U_j = V_j - \sum_{i=1}^j \frac{J}{J^2 + (\dot{\lambda} \alpha_j)^2} \left(\ddot{\lambda} \alpha_j + (\dot{\lambda})^2 \partial_\lambda \alpha_j \right). \quad [14]$$

As earlier, the tilde sign indicates that the right-hand side is gauge-equivalent to the CD Hamiltonian, and the two coincide only at $\dot{\lambda} = 0$. This equivalence is again very similar to the tilting of the tray in the waiter example. It is easy to see that the CD Hamiltonian has two distinct limits. At small velocities $|dV_j/dt| = |\dot{\lambda} v_j| \ll J^2$, the renormalization of hopping is negligible, and the CD term is a correction to the potential proportional to the acceleration $\ddot{\lambda}$ exactly like in the waiter example. Conversely, in the high-velocity limit, renormalization of the potential is negligible, and the CD Hamiltonian contains the renormalized hopping, which scales linearly with the velocity. This local hopping renormalization plays a role similar to the refractive index by locally changing the group velocity of particles in a way that essentially traps scattered particles.

Because the CD Hamiltonian depends only on the velocity and acceleration $\dot{\lambda}$ and $\ddot{\lambda}$, respectively, it is convenient (although not necessary) to deal with $\lambda(t)$, which has vanishing first and second derivatives at the beginning and the end of the protocol, such that $H_{CD} = H_0$ at these points. An example of such a protocol that we will use in this work is

$$\lambda(t) = \lambda_0 + (\lambda_f - \lambda_0) \sin^2 \left(\frac{\pi}{2} \sin^2 \left(\frac{\pi t}{\tau} \right) \right), \quad t \in (0, \tau). \quad [15]$$

This protocol ramps $\lambda(t)$ from the initial value λ_0 to the final value λ_f during the time τ .

Uniform linear potential. In the case of a general time-dependent force $V_j(\lambda) = \lambda_j$, with λ playing the role of an effective electric field, the solution of Eq. 11 is very simple: $\alpha_j = -J/\lambda^2$. It is easy to check that this solution is, in fact, exact up to the boundary terms (details are in *SI Text, Fig. S1*) (i.e., $\mathcal{A}_\lambda^* = \mathcal{A}_\lambda$). Remarkably, because α is constant, the effective potential U_j remains linear. Additionally, the effective hopping is constant across the lattice, which allows us to absorb the hopping renormalization into the timescale, because one can always rescale the Hamiltonian by an arbitrary factor without exciting the system.

As a result of these transformations, the real CD Hamiltonian is structurally the same as the naive Hamiltonian. One simply has to switch on the electric field in a different way to avoid ending in an excited state; that is, for each protocol λ , the CD protocol is

$$\lambda_{CD} = \frac{\lambda}{\sqrt{1 + \mu \ddot{\lambda}}} \left(1 - \frac{\mu \ddot{\lambda}}{1 + \mu \ddot{\lambda}} \right), \quad \text{where } \mu = 1/\lambda. \quad [16]$$

This protocol is illustrated in Fig. 2 for $\lambda_0 = 0.1J$ and $\lambda_f = J$ (blue line in Fig. 2) for a particular choice $\tau = 5/J$. To allow for enough transport of particles, there is an initial pulse in the field with an amplitude that is significantly larger than the final one.

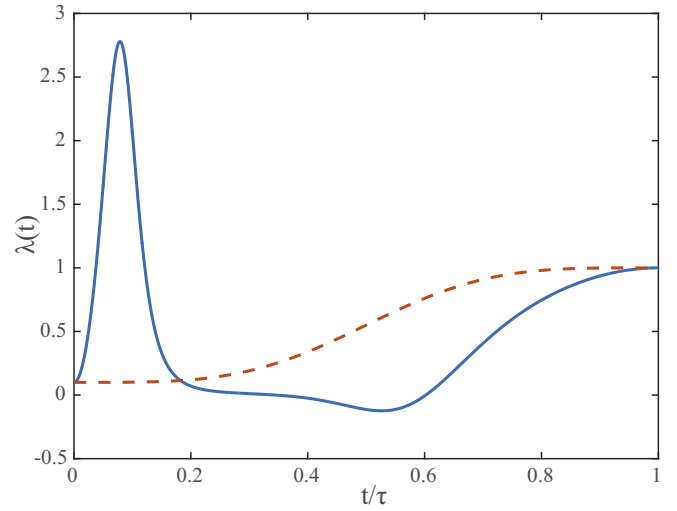


Fig. 2. CD protocol. Example of a naive adiabatic protocol (Eq. 15) to switch on a linear potential from $\lambda_0 = 0.1J$ to a final value of $\lambda_f = J$ without exciting the system (dashed red line) and a CD protocol that does exactly that in a time $\tau = 5/J$ (blue line).

After this pulse, the field is much flatter and first goes opposite to the target field before reversing to the right direction to reach the final desired field λ_f . The naive and the CD protocols come closer to each other as one increases the duration τ , although they always significantly differ at small fields.

Let us also comment on the opposite limit of instantaneous protocol, $\tau \rightarrow 0$, where the bare coupling turns on as a step-like function: $\lambda(t) \rightarrow \lambda_0 + (\lambda_f - \lambda_0)\theta(t)$. In this case, the Galilean term dominates the CD Hamiltonian. It is then convenient to formally use the chain rule and parametrize time in terms of the coupling $i\partial_t \psi = i\dot{\lambda} \partial_\lambda \psi$, such that the Schrödinger equation reads

$$i\partial_\lambda |\psi\rangle = \mathcal{A}_\lambda^* |\psi\rangle. \quad [17]$$

If the gauge potential is exact, then this equation describes the fastest route to perform adiabatic evolution. In particular, a slow turning on of the uniform field is equivalent to dynamics governed by the Hamiltonian

$$\mathcal{A}_\lambda^* = -\frac{iJ}{\lambda^2} \sum_j \left(c_{j+1}^\dagger c_j - c_j^\dagger c_{j+1} \right). \quad [18]$$

Because the coupling λ effectively plays the role of time, we see that the total time required to load the system into the ground state according to Eq. 17 diverges as the initial or final electric field approaches zero. As we will discuss elsewhere, this divergence is fundamentally related to the divergence of the quantum speed limit. Let us point out that, at $\tau \rightarrow 0$, there is no smooth real CD protocol, because it contains the acceleration terms that become singular.

Fighting Anderson orthogonality: Inserting a potential. Another somewhat more involved problem is the adiabatic insertion of a scattering potential into the Fermi gas. This problem is harder than it might seem. The difficulty can be understood from the perspective of Anderson's orthogonality catastrophe (20), which states that the ground state of the homogeneous Fermi gas and the gas with a single impurity are orthogonal in the thermodynamic limit. In addition, the system is gapless; as a consequence, standard arguments exploited in the adiabatic quantum computing literature (21, 22) suggest that, to load the potential adiabatically, one has to scale the ramp velocity to zero with the inverse system size. We checked that this is indeed the case for the naive

loading protocol. The situation changes dramatically with the CD driving.

To obtain the gauge potential, we numerically solve Eq. 11 in a box of size L with vanishing boundary conditions $\alpha_{j=-L/2} = \alpha_{j=L/2} = 0$. In Fig. 3, we show time dependence of the squared fidelity of the wave function and the instantaneous ground state: $F^2(t) = |\langle \psi(t) | \psi_{GS}(t) \rangle|^2$ for different protocols. The total system size is $L = 512$ at half-filling (256 particles). The system is initially prepared in the ground state at zero potential, and then, we are turning on a repulsive Eckart potential of the form

$$V_j(\lambda) = \frac{\lambda}{\cosh^2 j/\xi}, \quad j \in [-L/2, L/2].$$

We choose $\xi = 8$ and turn on $\lambda(t)$ according to the protocol (Eq. 15) with $\lambda_0 = 0$ and $\lambda_f = 2J$. The total duration of the protocol is $\tau = 10/J$. The naive protocol indeed fails completely, giving the very small final fidelity $F^2(\tau) \approx 2 \cdot 10^{-19}$ as expected. This result is only marginally better than the fidelity of the initial state and the final state, which is $5 \cdot 10^{-20}$. The CD protocol, however, gives fidelity of the order of $1/2$, gaining more than 18 orders of magnitude. This value implies a 50% chance of preparing the system in the exact many-body ground state. Notice that, although the imaginary CD protocol (purple line in Fig. 3) keeps instantaneous fidelity high at all times, for the real protocol, exactly like in the waiter case, the instantaneous fidelity at intermediate times drops to a very small value. High fidelity is only recovered at the end of the protocol, where the velocity λ becomes close to zero. For additional details on the performance see Figs. S2–S6.

1D Spin Chain. In the previous examples, we focused on free particle 1D systems, where an exact solution for $\mathcal{A}_\lambda$ always exists, and the variational approach merely helps one to find a simpler and easier way to implement local approximate $\mathcal{A}_\lambda^*$. Many-particle systems are intrinsically chaotic, and as we already discussed, the exact gauge potential simply does not exist in the form of a local operator. In such cases, approximate methods for finding $\mathcal{A}_\lambda^*$ are simply required to find CD protocols. To illustrate the power of the variational approach, let us consider an Ising spin chain in the presence of a transverse and longitudinal field.

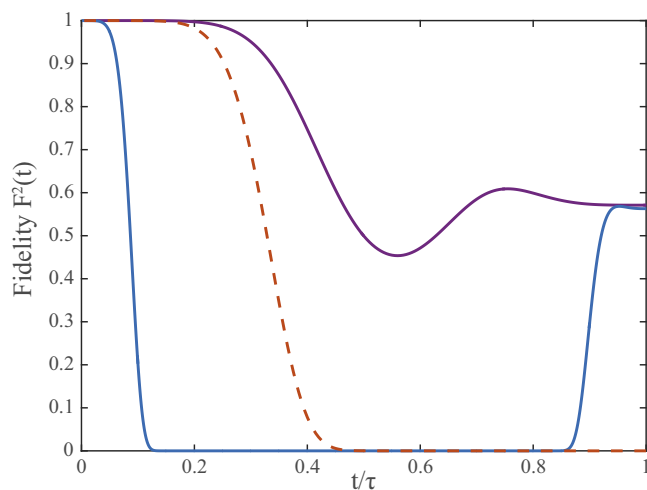


Fig. 3. Inserting local potential. The probability to be in the adiabatic ground state when inserting a scattering potential quickly decays to a small value for the naive protocol (dashed red line). By CD driving with a local complex gauge, the state stays much closer to the ground state, and a final fidelity of about $1/2$ is reached (purple line). A gauge-equivalent real Hamiltonian, with renormalized hopping and potential, results in the same final fidelity but is almost orthogonal to the ground state at intermediate times (blue line).

This system is one of the simplest nonintegrable models with very rich phase diagram (23, 24). The Hamiltonian of this system reads

$$H_0 = \sum_j (J\sigma_j^z\sigma_{j+1}^z + Z_j\sigma_j^z + X_j\sigma_j^x), \quad [19]$$

where σ_j^z and σ_j^x are the Pauli matrices. We allow all couplings to depend on some tuning parameter λ , which in turn, depends on time. The simplest gauge potential, which is purely imaginary, is the magnetic field along the y direction:

$$\mathcal{A}_\lambda^* = \sum_j \alpha_j \sigma_j^y. \quad [20]$$

Recall that $G_\lambda \equiv \partial_\lambda H + i[\mathcal{A}_\lambda^*, H]$; hence,

$$G_\lambda = \sum_j ((X'_j - 2Z_j\alpha_j)\sigma_j^x + 2\alpha_j J(\sigma_j^x\sigma_{j+1}^z + \sigma_j^z\sigma_{j+1}^x)) + \sum_j ((Z'_j + 2X_j\alpha_j)\sigma_j^z + J'\sigma_j^z\sigma_{j+1}^z), \quad [21]$$

where prime stands for the derivative with respect to λ . Because Pauli matrices are traceless, computing the Hilbert–Schmidt norm of this operator is trivial and amounts to adding up squares of coefficients in front of independent spin terms:

$$2^{-L}\text{Tr}(G_\lambda^2) = \sum_j ((X'_j - 2Z_j\alpha_j)^2 + 8\alpha_j^2 J^2) + \sum_j ((Z'_j + 2X_j\alpha_j)^2 + (J')^2). \quad [22]$$

Minimizing the quadratic form with respect to α_j immediately yields the optimal variational solution:

$$\alpha_j = \frac{1}{2} \frac{Z_j X'_j - X_j Z'_j}{Z_j^2 + X_j^2 + 2J^2}. \quad [23]$$

Note that the y magnetic field is strictly local, because it only depends on the local values of the x and z magnetic fields. For $J = 0$, this gauge potential is exact, because it is simply a generator of local spin rotations in the $x - z$ plane. Note that $\mathcal{A}_\lambda^*$ vanishes if either $h_x = 0$ or $h_z = 0$, implying that the leading contribution to $\mathcal{A}_\lambda$ in this case actually comes from two-spin terms. To second order, we can include two-spin terms into the variational ansatz:

$$\mathcal{A}_\lambda^* = \sum_j [\alpha_j \sigma_j^y + \beta_j (\sigma_j^y \sigma_{j+1}^z + \sigma_j^z \sigma_{j+1}^y)] + \sum_j \gamma_j (\sigma_j^y \sigma_{j+1}^x + \sigma_j^x \sigma_{j+1}^y). \quad [24]$$

The coefficients $\alpha_j, \beta_j, \gamma_j$ can directly be found by minimizing the quadratic action (Eq. 8), resulting in a linear set of coupled equations, which can be easily solved numerically. This variational solution dramatically enhances performance of the annealing protocol, loading spins from an initial product state to the ground state of the Hamiltonian (Eq. 19) (SI Text; specific protocols are analyzed in Figs. S7–S9).

Following the orthogonality catastrophe example, let us consider a CD protocol for turning on a local magnetic field. Specifically, we consider turning on an additional x magnetic field λ from 0 to the final value $\lambda_f = -10J$ in a periodic chain described by the Hamiltonian $H_0 + \lambda\sigma_0^x$, where H_0 is given by Eq. 19 with $J = 1$, $Z_j = 2$, and $X_j = 0.8$. We compare three protocols: bare protocol with no CD driving, CD protocol with only local CD driving (Eq. 20), and the optimal two-spin CD protocol (Eq. 24). We verified that the variational coefficients $\alpha_j, \beta_j, \gamma_j$ rapidly (exponentially) decay with j away from the site $j = 0$, and therefore, the CD protocol effectively remains local. Note that we can find the CD protocol in the thermodynamic limit;

however, we have to apply it to a finite chain to verify its performance. The probability of recovering the system in the ground state for a chain of 15 spins is shown in Fig. 4. For fast protocols, there is a significant reduction in excess energy and a corresponding increase in fidelity for the CD protocols. In particular, in the instantaneous quench limit $\tau \rightarrow 0$, the CD driving gives almost a factor of 10 gain in fidelity and a similar reduction in the heating. Interestingly, in this limit, dynamics is entirely governed by $\mathcal{A}_\lambda^*$, which is exponentially localized near $j = 0$. Loading into a strong magnetic field can be also viewed as changing the boundary conditions in the system; effectively, it cuts the chain in two pieces. Moreover, because of time-reversal symmetry, the fidelity of joining the chain back together is identical. Therefore, in this case, $\mathcal{A}_\lambda^*$ can be interpreted as an effective boundary Hamiltonian generating adiabatic boundary transformations on the ground-state wave function.

Next, let us use the same example to analyze another application of the CD driving, namely suppression of dissipation in a noisy system. Let us now assume that the site $j = 0$ is subject to a small white noise in the magnetic field in the x direction. Physically, this noise can stem from coupling of this site to a nearby impurity or a quantum dot. A convenient measure of dissipation in the system is the rate of spread of energy fluctuations $d_t \delta_E^2$. If we implement exact CD protocol, then the system follows instantaneous eigenstates, and the energy fluctuations remain constant in time, so that the rate is zero. At finite temperatures, $d_t \delta_E^2$ is related to the usual heating rate dE/dt by the fluctuation dissipation relation (19). However, the spread of energy fluctuations has a well-defined limit even for the infinite temperature states, where the heating rate vanishes. Within Fermi's golden rule and under the assumption that the spectral density of the fluctuating field is white, the rate at which the fluctuations will increase if we initialize the system in a pure state $|n\rangle$ takes the following form:

$$\partial_t \delta_E^2 = S_{\lambda\lambda} \langle n | ([H_0, G_\lambda])^2 | n \rangle, \quad [25]$$

where $S_{\lambda\lambda}$ is the power spectral density of the noise (i.e., the inverse bandwidth of the noise). Fig. 5 shows the normalized production rate $\partial_t \delta_E^2 / S_{\lambda\lambda}$ for every eigenstate of an ergodic chain of 15 spins. We clearly see a reduction in fluctuations across the entire spectrum, comparable with the reduction in excess energy for the ground-state protocol discussed above. Interestingly, CD driving not only reduces dissipation but also, reduces its fluctuations between different eigenstates.

To understand better the performance of the CD protocol, we can analyze the spectral decomposition of the dissipation as a function of the absorption frequency. Specifically, we look at the

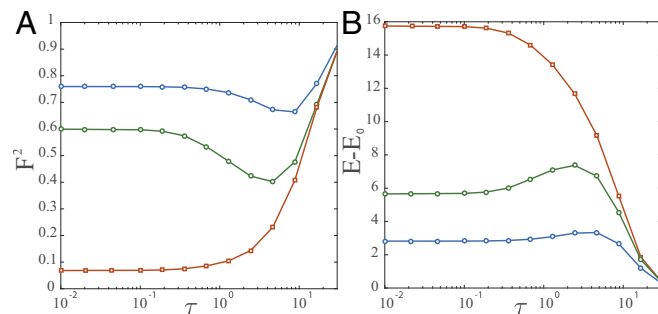


Fig. 4. Local spin flip. A single spin in a chain of 15 spins is aligned with the x direction by switching a strong local magnetic field in the x direction (details are in the text). A shows the fidelity to recover the system in the ground state after a protocol of duration τ . B shows the final energy above the ground state. The red squares are associated with the naive protocol, the green circles are associated with strictly local CD driving, and the blue circles are associated with CD driving (Eq. 24).

lifetime of a state Γ_n within Fermi's golden rule when we subject the system to a periodic drive with frequency ω :

$$\Gamma_n(\omega) = S_{\lambda\lambda}(\omega) \sum_m |\langle m | G_\lambda | n \rangle|^2 \delta(E_m - E_n - \omega), \quad [26]$$

where the δ function is broadened, such that it contains several eigenstates leading to a smooth dependence of $\Gamma_n(\omega)$. The result is shown in Fig. 6. All three protocols show a very narrow peak in the transition rate around $\omega = 0$ followed by a much broader distribution at large frequencies. The small frequency transitions represent mixing between nearby eigenstates. They lead to very small dissipation and are not affected by the CD driving terms. Conversely, the high-frequency transitions leading to large energy transfer from the noise to the system are strongly suppressed by the CD driving.

Classical Spin Chain. Let us briefly show how the developed ideas can be applied to classical systems. Specifically, we will consider a classical rotor model with the Hamiltonian

$$H_0 = J \sum_j \vec{S}_j \vec{S}_{j+1} + X_0(\lambda) S_0^x + Z_0(\lambda) S_0^z, \quad [27]$$

where $\vec{S}_j$ are 3D classical angular momenta. Because $\vec{S}_j^2 = S^2$ is conserved under the Hamiltonian dynamics, we will set $S^2 = 1$. This Hamiltonian, with appropriate rescaling of the coupling constants, can be obtained as a large spin limit of a quantum Heisenberg spin chain with an additional local magnetic field. This Hamiltonian is a direct analog of a quantum spin model analyzed earlier (Eq. 19), except that we use isotropic Heisenberg coupling. Because the Heisenberg model is not integrable, there is no need to introduce extra static magnetic fields. As in the quantum case in the leading order, we will seek the gauge potential in the form

$$\mathcal{A}_\lambda^* = \alpha_0(\lambda) S_0^y. \quad [28]$$

Recall that the Poisson bracket between any two functions can be expressed as

$$\left\{ A(\vec{S}_j), B(\vec{S}_j) \right\} = \sum_j \epsilon_{abc} S_j^a \frac{\partial A}{\partial S_j^b} \frac{\partial B}{\partial S_j^c},$$

where $a, b, c = \{x, y, z\}$, and ϵ_{abc} is a fully antisymmetric tensor. Using this definition we find that

$$G_\lambda = \partial_\lambda H_0 + \{H_0, \mathcal{A}_\lambda^*\} = (X'_0 - \alpha Z_0) S_0^x + (Z'_0 + \alpha X_0) S_0^z + J\alpha (S_0^z (S_1^x + S_{-1}^x) - S_0^x (S_1^z + S_{-1}^z)). \quad [29]$$

From this expression, we easily obtain $\|G_\lambda\|^2$ by integrating the square of G_λ over spin directions:

$$\|G_\lambda\|^2 = \frac{1}{3} (X'_0 - \alpha Z_0)^2 + \frac{1}{3} (Z'_0 + \alpha X_0)^2 + \frac{4}{9} J^2 \alpha^2. \quad [30]$$

Minimizing this with respect to α , we find the optimum solution:

$$\alpha = \frac{Z_0 X'_0 - X_0 Z'_0}{X_0^2 + Z_0^2 + \frac{4}{3} J^2}. \quad [31]$$

This expression is very similar to the quantum result (Eq. 23) with a slightly different prefactor in the J^2 . (The overall factor $1/2$ in front is just because of the difference between spins and Pauli matrices.) For the classical model with only $z-z$ interactions, the only difference in α would be in a prefactor in front of J^2 : $2/3$ instead of $4/3$. In a similar fashion, one can extend the variational ansatz to higher orders, including various terms odd in powers of S^y like $S_j^y S_{j+1}^{(z,x)}$.

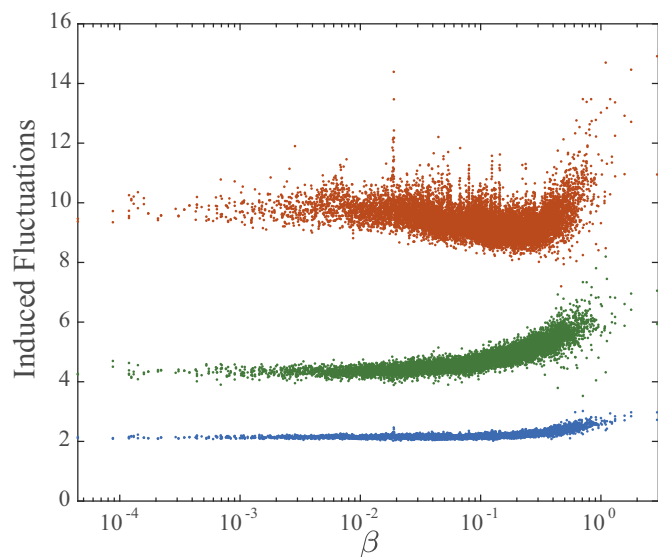


Fig. 5. Spin chain fluctuations. Normalized energy fluctuations production rate in a chain of $L = 15$ spins with $X = 0.9$, $Z = 0.8$, and $J = 1$ when a single spin is subject to a random weak magnetic field in the x direction. The effective inverse temperature of each eigenstate β is defined to match its energy in an equivalent thermal ensemble. The red, green, and blue dots represent results for no CD term, the best single-site CD term, and the best two-site CD term, respectively.

Methods

Consider a state $|\psi\rangle$ evolving under a time-dependent Hamiltonian $H_0(\lambda(t))$:

$$i\hbar\partial_t|\psi\rangle = H_0(\lambda(t))|\psi\rangle, \quad [32]$$

where the full time dependence of the Hamiltonian is caused by an external drive $\lambda(t)$. Let us go to the rotating frame where the Hamiltonian remains stationary (diagonal at all times). This diagonalization can be always achieved by a λ -dependent unitary transformation $U(\lambda(t))$, which also expresses the state in the instantaneous eigenbasis of the Hamiltonian. It is easy to see that the wave function in the moving frame $|\tilde{\psi}\rangle = U(\lambda)|\psi\rangle$ satisfies the effective Schrödinger equation (17):

$$i\hbar\partial_t|\tilde{\psi}\rangle = (\tilde{H}_0(\lambda(t)) - \dot{\lambda}\tilde{\mathcal{A}}_\lambda)|\tilde{\psi}\rangle, \quad [33]$$

where $\tilde{H}_0$ is the Hamiltonian in the instantaneous basis, and $\tilde{\mathcal{A}}_\lambda$ is the adiabatic gauge potential in the moving frame:

$$\begin{aligned} \tilde{H}_0(\lambda(t)) &= U^\dagger H_0(\lambda(t))U = \sum_n \epsilon_n(\lambda)|n\rangle\langle n|, \\ \tilde{\mathcal{A}}_\lambda &= U^\dagger i\partial_\lambda U. \end{aligned} \quad [34]$$

Differentiating the relation $\tilde{H}_0 = U^\dagger H_0 U$ with respect to λ and using the fact that $\partial_\lambda \tilde{H}_0$ commutes with $\tilde{H}_0$, one can check that the adiabatic gauge potential satisfies the following equation (17):

$$i\hbar(\partial_\lambda H_0 + F_{\text{ad}}) = [\mathcal{A}_\lambda, H_0], \quad [35]$$

where F_{ad} is the adiabatic or generalized force operator:

$$F_{\text{ad}} = -\sum_n \partial_\lambda \epsilon_n(\lambda) |n(\lambda)\rangle \langle n(\lambda)|, \quad [36]$$

and $\mathcal{A}_\lambda = U\tilde{\mathcal{A}}_\lambda U^\dagger = i(\partial_\lambda U)U^\dagger$ is the adiabatic gauge potential in the laboratory frame. Although we used the moving frame to derive Eq. 35, it is an operator equation, which is valid in any frame, including the laboratory frame. Note that Eq. 2 trivially follows from Eq. 35, because F_{ad} by construction commutes with the Hamiltonian.

Now let us discuss in more detail how Eq. 35 can be reformulated as the minimization problem leading to the variational ansatz. For this purpose, let us choose some trial gauge potential $\mathcal{A}_\lambda^*$ and define an operator G_λ :

$$G_\lambda = \partial_\lambda H_0 + \frac{i}{\hbar}[\mathcal{A}_\lambda^*, H_0].$$

This operator also has a well-defined classical limit. It is clear from Eq. 35 that, for $\mathcal{A}_\lambda^* = \mathcal{A}_\lambda$, we have $G_\lambda = -F_{\text{ad}}$. The diagonal elements of G_λ in the

basis of the Hamiltonian do not depend on $\mathcal{A}_\lambda^*$: $\langle n|G_\lambda|n\rangle = \partial_\lambda \epsilon_n(\lambda)$. Thus, different choices of $\mathcal{A}_\lambda^*$ only affect the off-diagonal elements of G_λ . For the true gauge potential $\mathcal{A}_\lambda$, G_λ has no off-diagonal elements; it thus corresponds to the operator G_λ with minimal Hilbert–Schmidt norm. Formally, this equivalence can be seen from the distance between G_λ and $-F_{\text{ad}}$:

$$\begin{aligned} \mathcal{D}(\mathcal{A}_\lambda^*) &= \text{Tr}[(G_\lambda + F_{\text{ad}})^2] \\ &= \text{Tr}\left[\left(\partial_\lambda H_0 + F_{\text{ad}} + \frac{i}{\hbar}[\mathcal{A}_\lambda^*, H_0]\right)^2\right]. \end{aligned} \quad [37]$$

Using cyclic properties of the trace, this distance becomes

$$\mathcal{D}(\mathcal{A}_\lambda^*) = \mathcal{S}(\mathcal{A}_\lambda^*) - \text{Tr}[F_{\text{ad}}^2], \quad [38]$$

with the action $\mathcal{S}(\mathcal{A}_\lambda^*)$ given by Eq. 8. Minimizing the distance with respect to $\mathcal{A}_\lambda^*$ results in Eq. 2. We would like to stress that an enormous gain has been made by moving from the original Eq. 35 to Eq. 2, because the adiabatic force has been eliminated. In fact, neither the action (Eq. 8) nor the Euler–Lagrange Eq. 2 make any reference to the adiabatic force, which is generally hard to compute. Nonetheless, by exactly minimizing (Eq. 8), one recovers the exact adiabatic gauge potential $\mathcal{A}_\lambda = i(\partial_\lambda U)U^\dagger$, which implies that the matrix elements of $\mathcal{A}_\lambda$ between instantaneous eigenstates satisfy $\langle m(\lambda)|\mathcal{A}_\lambda|n(\lambda)\rangle = \langle m(\lambda)|i\partial_\lambda|n(\lambda)\rangle$. Consequently, this CD drive reproduces the full adiabatic evolution, including the Berry phase. Approximate gauge potentials can, therefore, be used to extract an approximate Berry curvature and phase.

Discussion and Conclusion

Building on the concept of transitionless driving, we have developed a variational principle that allows one to construct approximate variational CD protocols. Using this variational ansatz, we obtained best local CD protocols and showed that they can strongly decrease heating of highly excited states and increase fidelity of the ground-state preparation by many orders of magnitude. Efficient CD protocols can find many different applications from constructing fast and efficient annealing protocols for both quantum computers and quantum simulators to engineering thermodynamic engines operating close to the maximum efficiency at fast speeds. A key advantage of the variational method is that it allows one to find such protocols without the need for knowing any details about the spectrum of the Hamiltonian or the structure of its eigenstates, which are usually very hard to obtain.

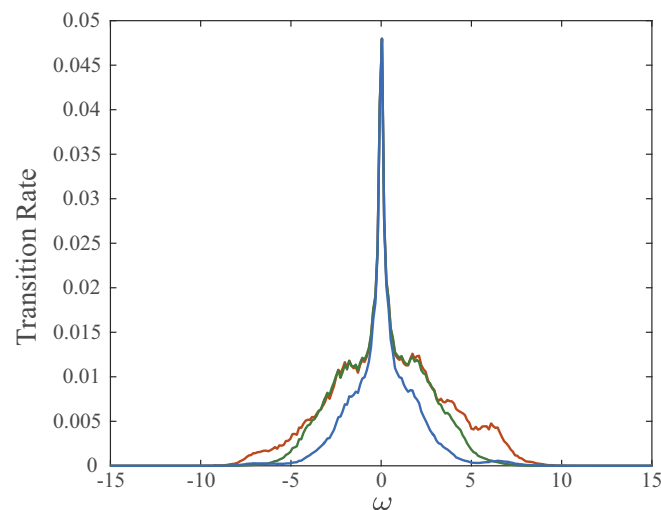


Fig. 6. Spin chain transitions. Average normalized transition rate over states with effective temperature $\beta = 0.1$. The parameters and the colors are the same as in Fig. 5.

We illustrated the ideas using various examples: (i) inserting (and moving) a local potential barrier into a Fermi sea, effectively fighting the Anderson orthogonality catastrophe, and (ii) locally flipping a spin on an ergodic quantum spin chain. In all cases, we showed that CD causes a dramatic decrease of heating and improvements in the final fidelity even in the gapless regimes, where standard arguments based on the gap indicate that the high-fidelity state preparation is not possible at these fast rates. It remains to be seen how efficient and robust the

CD protocol can be in more complicated situations, such as crossing phase transitions or annealing systems with slow glassy dynamics.

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